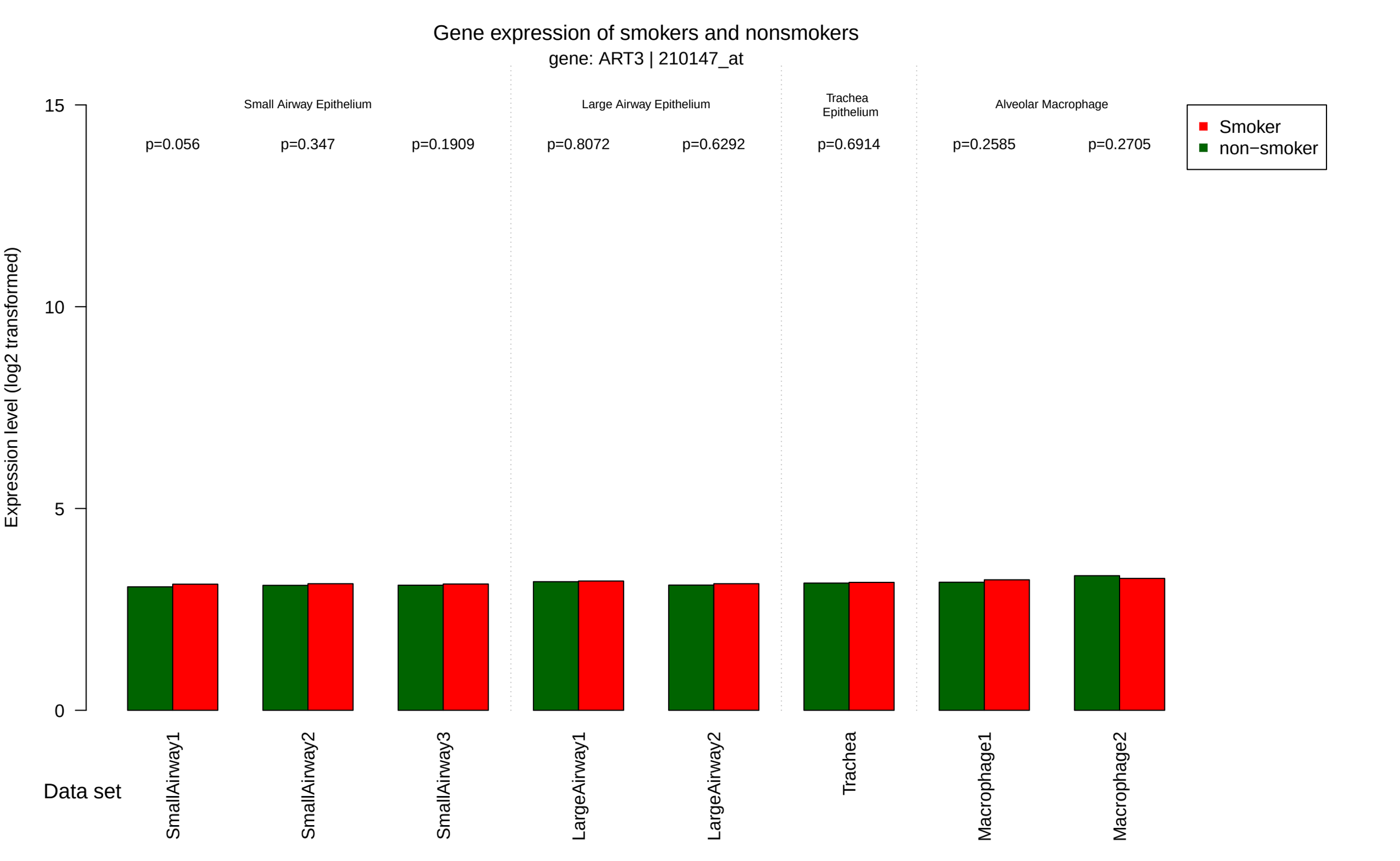
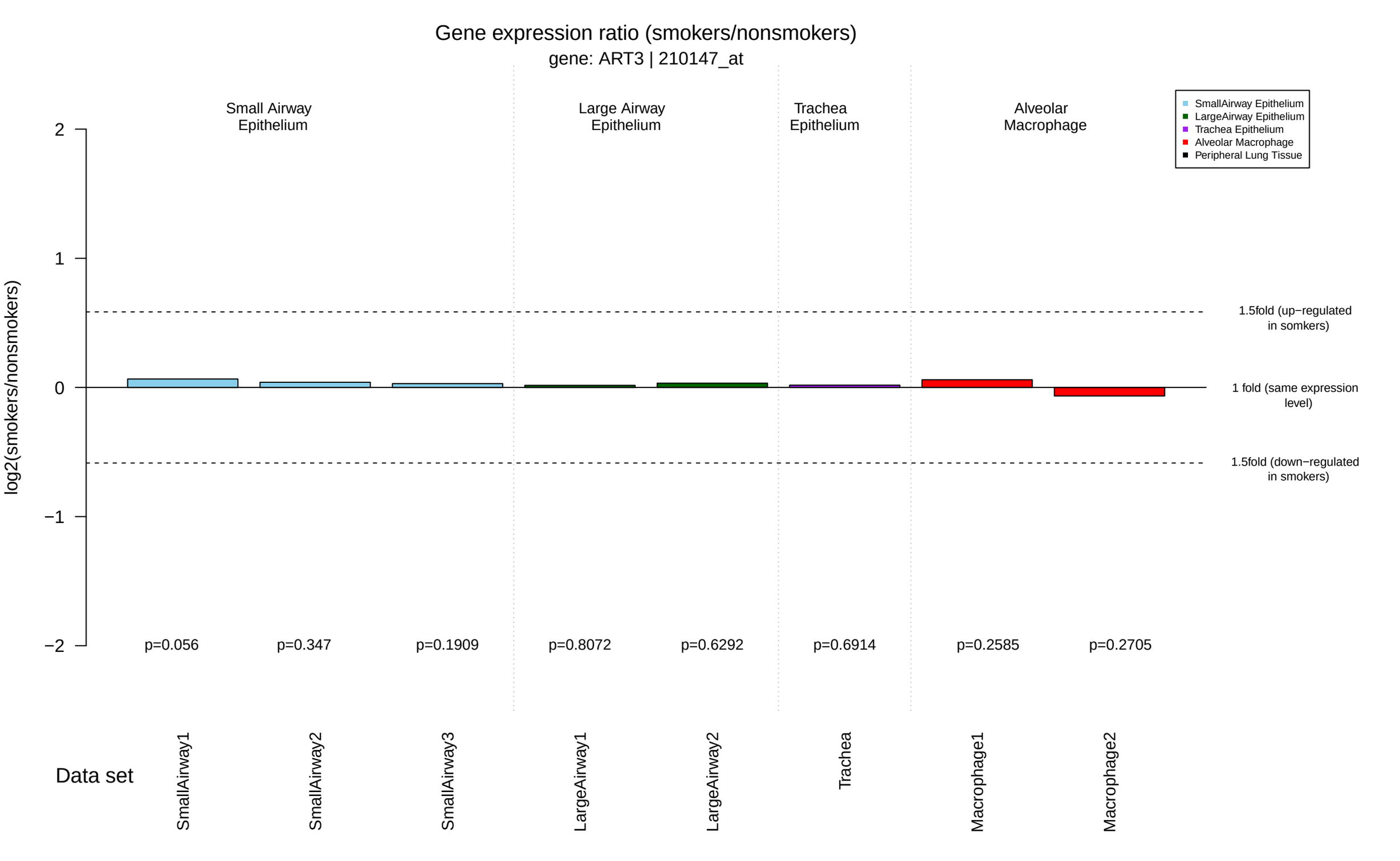
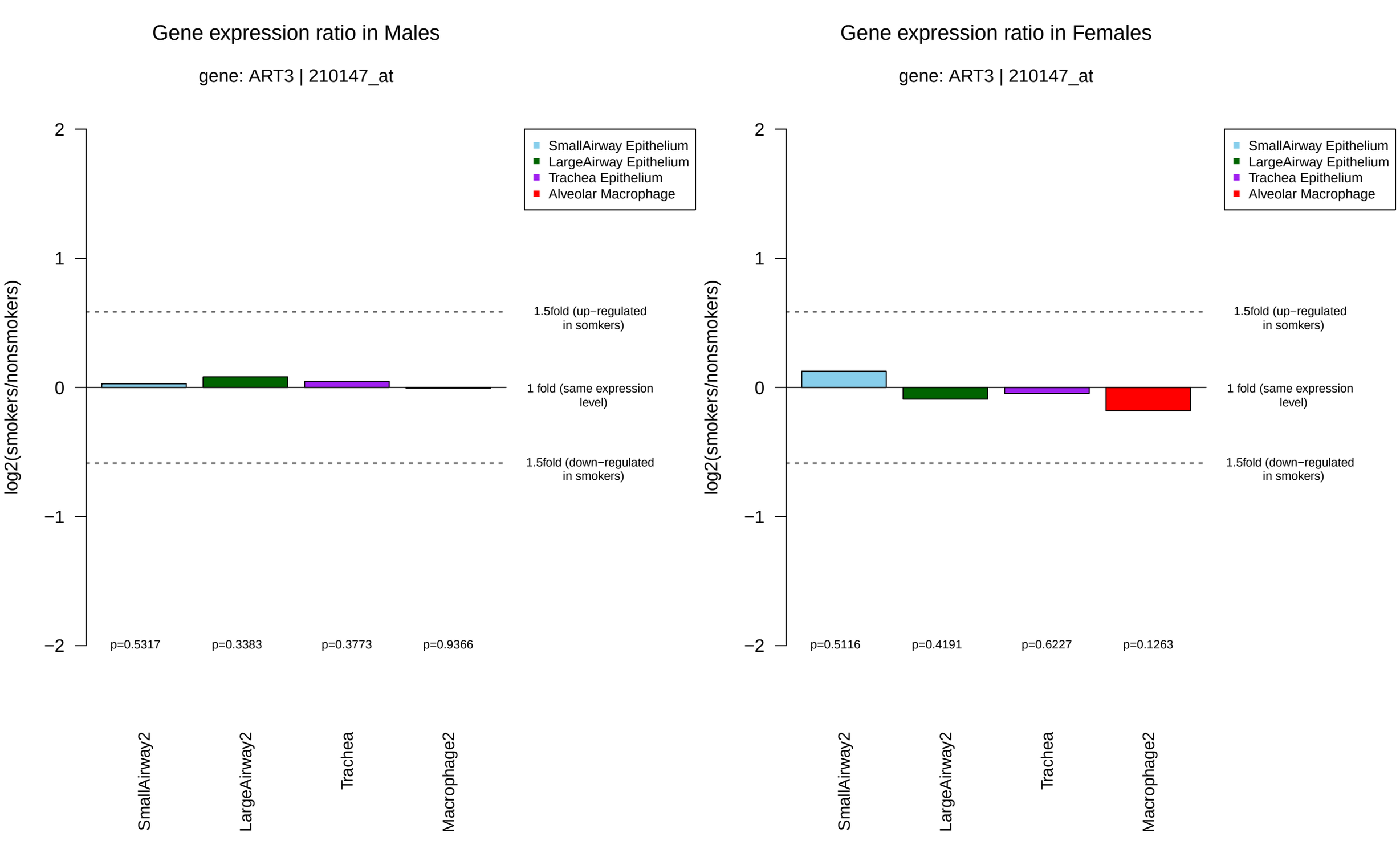
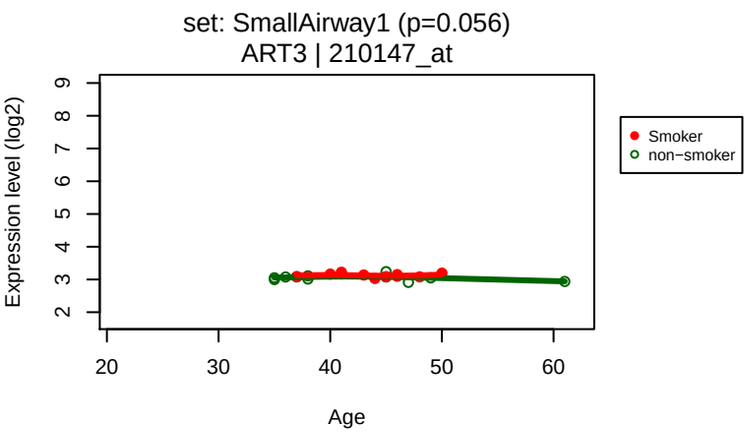
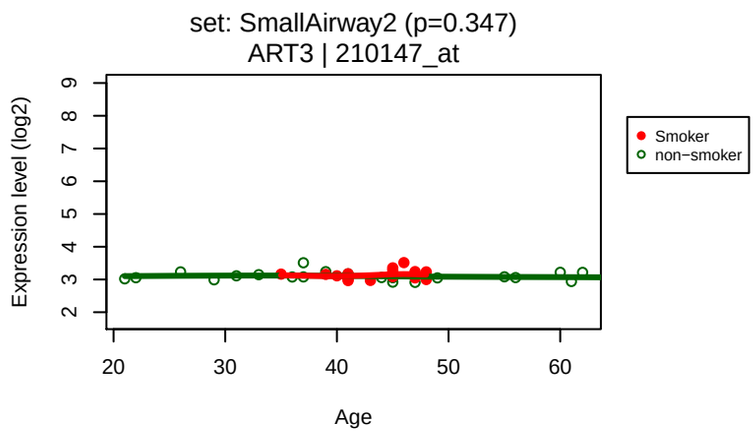
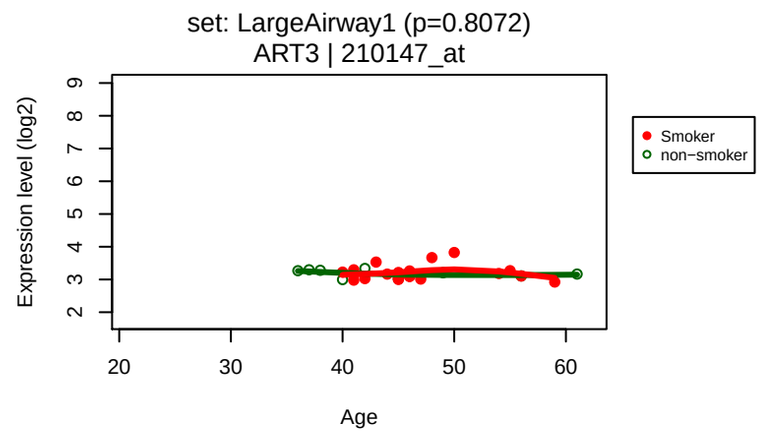

Gene expression of smokers and nonsmokers
gene: ART3 | 210147_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.056
p=0.347
p=0.1909
p=0.8072
p=0.6292
p=0.6914
p=0.2585
p=0.2705
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

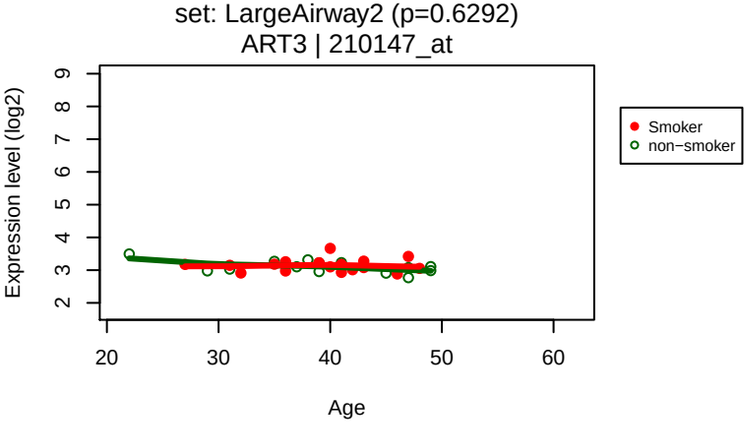

Gene expression ratio (smokers/nonsmokers)
gene: ART3 | 210147_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.056
p=0.347
p=0.1909
p=0.8072
p=0.6292
p=0.6914
p=0.2585
p=0.2705
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

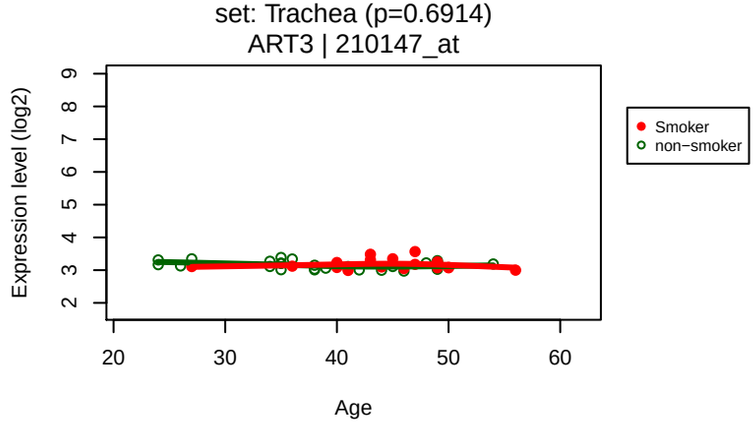

Gene expression ratio in Males
gene: ART3 | 210147_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5317
p=0.3383
p=0.3773
p=0.9366
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ART3 | 210147_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5116
p=0.4191
p=0.6227
p=0.1263
SmallAirway2
LargeAirway2
Trachea
Macrophage2

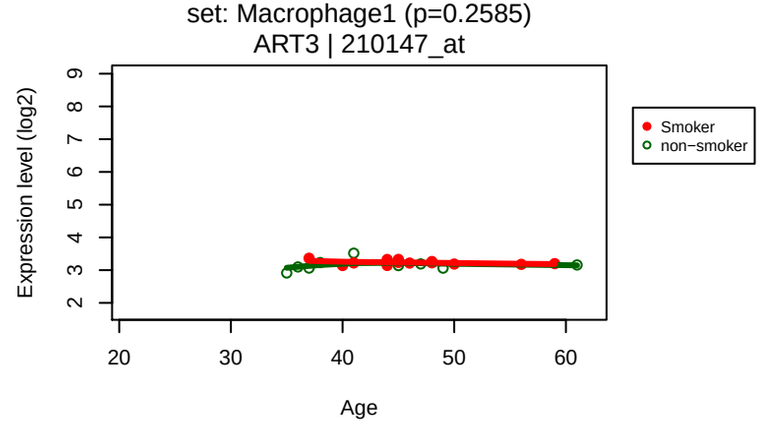

set: SmallAirway1 (p=0.056)
ART3 | 210147_at
Expression level (log2)
Age
Smoker
non−smoker

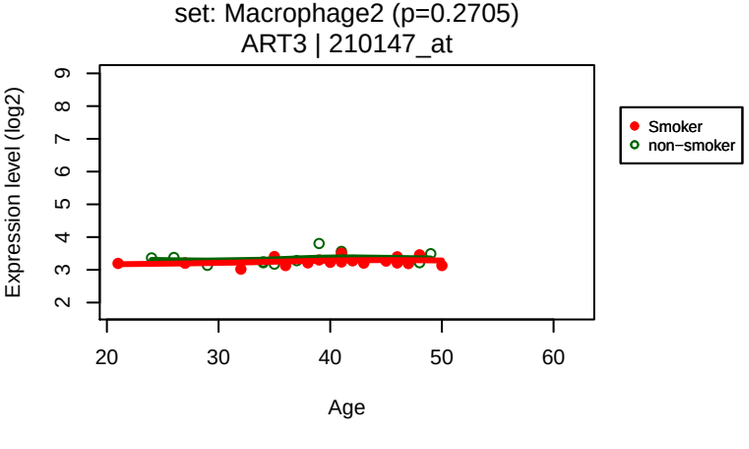

set: SmallAirway2 (p=0.347)
ART3 | 210147_at
Expression level (log2)
Age
Smoker
non−smoker

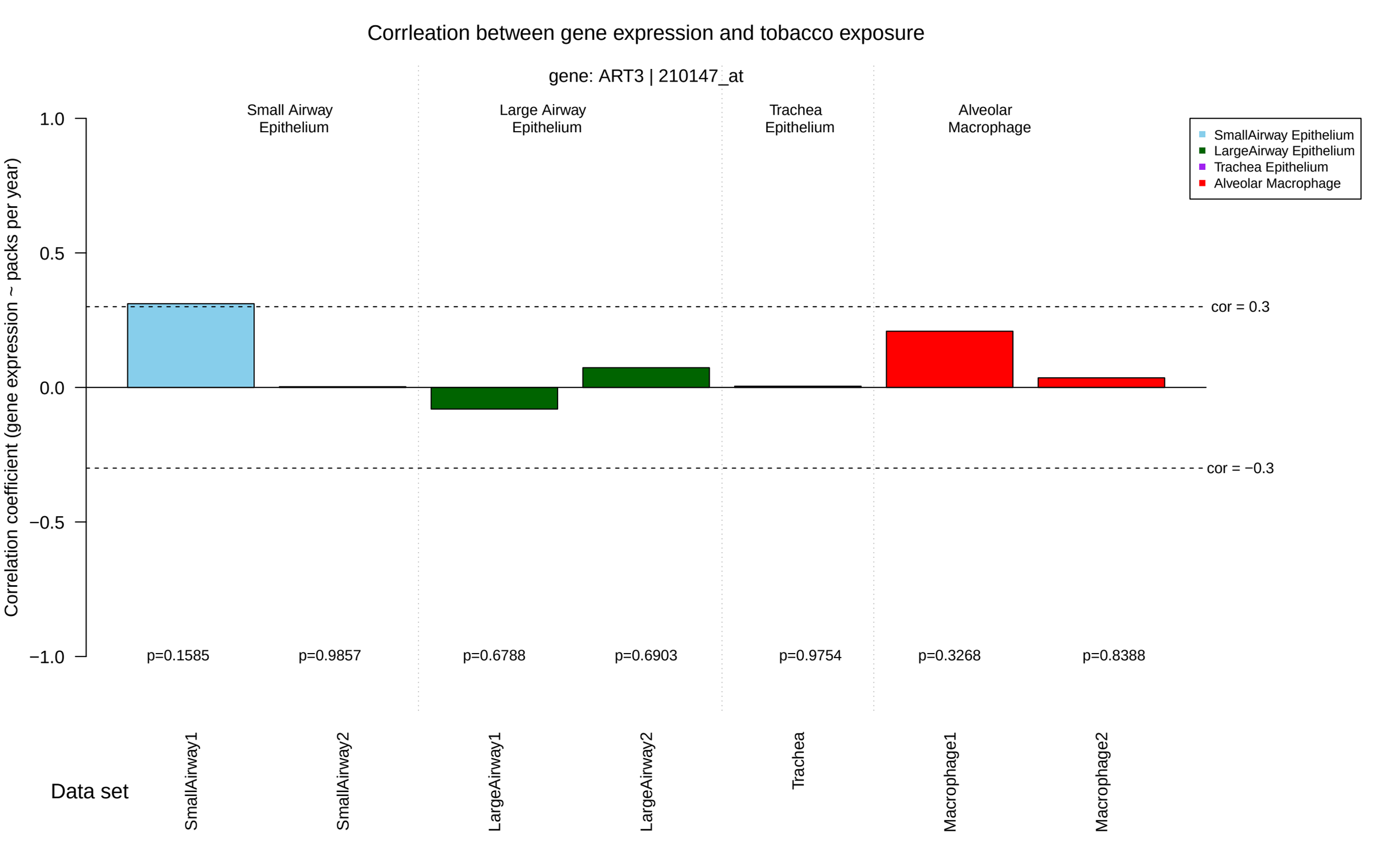

set: LargeAirway1 (p=0.8072)
ART3 | 210147_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6292)
ART3 | 210147_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6914)
ART3 | 210147_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2585)
ART3 | 210147_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2705)
ART3 | 210147_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ART3 | 210147_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.1585
p=0.9857
p=0.6788
p=0.6903
p=0.9754
p=0.3268
p=0.8388
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2